

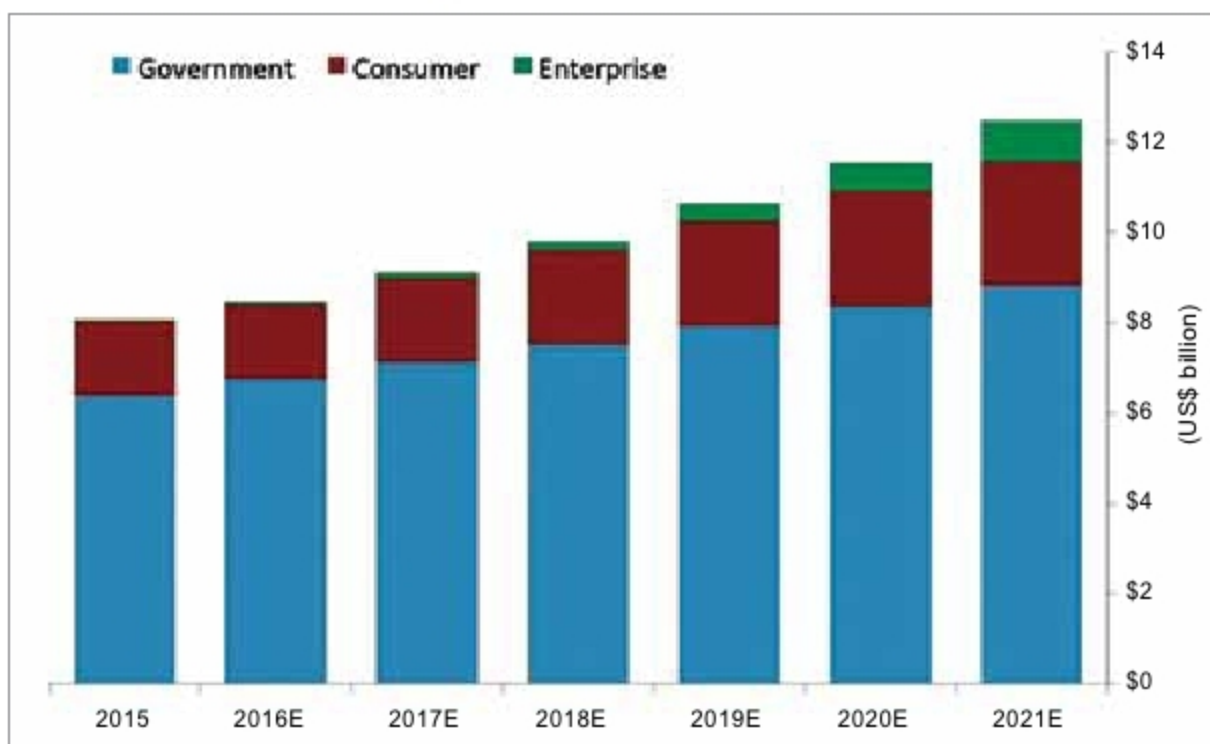


Fig. 3: Estimated global investment in drone hardware (Credit: BI Intelligence)

- Gathering information or supplying essentials for disaster management

- Thermal sensor drones for search and rescue operations

- Geographic mapping of inaccessible terrain and locations

- Precision crop monitoring

- Law enforcement and border control surveillance, and building safety inspections

- Storm tracking and forecasting hurricanes and tornadoes

- Surveillance by defence, police, large public sectors, security departments, and railways.

- Mapping large areas in land survey, surveying and mapping by Survey of India, mining industry, land records, etc

D'Souza adds, "Drones use significantly less manpower, thereby reducing long-term costs. Indian Railways is using UAVs for inspection and 3D mapping to bring to life their vision of a dedicated freight corridor. Heavy-duty drones offer much smoother management of inventory and have the ability to move the inventory easily from one warehouse to another. The upcoming biggest jump of technology will be the usage of drones in logistics. Nowadays, the biggest commercial usage of drones is the last-mile delivery with the usage of drones."

## Global drone market forecasts

According to a study by Market-sandmarkets, the drone logistics and transportation market is expected to

reach 11.20 billion dollars by 2022 and 29.06 billion dollars by 2027, at a compound annual growth rate (CAGR) of 21.01 per cent during the period. McKinsey predicts that AVs, including the delivery drones, will make up to 80 per cent of deliveries in the future, while NASA estimates that 2.6 million commercial drones will be in the air in 2020 itself.

According to a recent report by Goldman Sachs, the global drone market is projected to be a 100 billion dollars opportunity by 2020. Key market spending will include: military—70 billion dollars, consumer—17 billion dollars, and commercial/civil—13 billion dollars, with the last offering the fastest growth opportunity.

According to Mordor Intelligence, the drone logistics and transportation market would register a CAGR of over 20 per cent during the period 2020-2025.

A recent study from IDC's Customer Insights & Analysis mentions that, worldwide drone spending was 9.3 billion dollars in 2018 and is expected to grow at a faster rate than the overall market with a five-year CAGR of 32.1 per cent. Throughout this period, enterprise drone solutions are expected to incur more than half of all the drone spending as more interests and demand appear from consumer drone solutions. Category wise, the largest demand of drones during this period will be from precision agriculture/crop scouting (86.1

per cent CAGR), emergency service (86.4 per cent CAGR), and dispensing pesticides and fertilizers (109.4 per cent CAGR).

According to BI Intelligence, the commercial and civilian drone market has been growing at a CAGR of 19 per cent between 2015 and 2020. That dwarfs the CAGR of 5 per cent for military drones, which is just a fraction of the overall drone market.

The Federal Aviation Administration (FAA) forecasts that sales of commercial drones will reach 2.7 million by 2020 end, while the total number of UAVs seen in the sky will be around seven million during the same period. Currently, the largest application of commercial drones is in the field of video and still photography. But, a report from Global Market Insights highlights that the global commercial drone market will reach 17 billion dollars by 2024.

## Growth of drones in Indian market

Currently, both the market and drone industry in India are at a very nascent stage. But, according to experts, their potential for growth in India is colossal. In India, between 2017-2023, the expected growth rate of the drone market would be eighteen per cent per annum, claims 6Wresearch.

D'Souza says, "In India, the commercial use of drones has been limited by policy paralysis. But now, the Ministry of Civil Aviation has come up with a policy around drones and their commercial use."

According to a study by Unearth-Insight, there are about 100 drone startups in India, 681 in the United States and 129 in China. The drone market is still nascent world over, but rapidly growing with game-changing innovations yet to come.

In terms of creating new jobs, drones offer huge potential for engineers, UAV operators, pilots, and data analysis and processing experts. The potential for a job could be both full-time and contract-based, also opening the doors of internships for many. In this regard, FICCI and EY clearly mention that the UAV market will touch 885.7 million dollars in India by 2021 while the global market size will be 21.47 billion dollars. **EFY**